Today

- More reflection and transmission
- Simple harmonic oscillator
- Tunneling

2.6 Step potential

Next consider a particle of energy $E > V_0$ approaching a step potential from the left in the $x < 0$ region with

$$V(x) = \begin{cases} 0, & x < 0 \\ V_0, & x > 0. \end{cases}$$

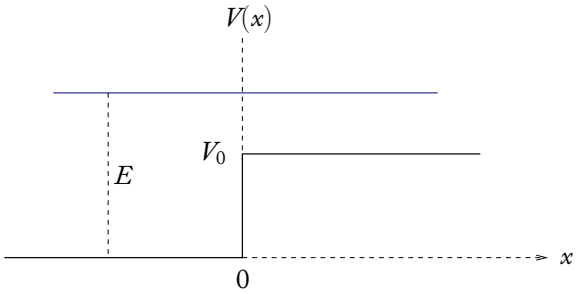


Figure 2.6.1: A particle with energy above a step potential.

Denoting the region $x < 0$ by I and that in $x > 0$ by II, and noting that there are both incident and reflected waves in region I while there is only transmitted wave in region II, we can immediately write the wavefunctions in each region,

$$\psi_I(x) = Ae^{ikx} + Be^{-ikx}, \qquad \psi_{II}(x) = Ce^{i\kappa x}, \tag{2.6.1}$$

where

$$k = \sqrt{\frac{2mE}{\hbar^2}}, \qquad \kappa = \sqrt{\frac{2m(E - V_0)}{\hbar^2}} \tag{2.6.2}$$

Matching $\psi(x)$ and $\psi'(x)$ across $x = 0$, we have

$$A + B = C, \qquad ik(A - B) = i\kappa C. \tag{2.6.3}$$

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Solving these simultaneous equations for B and C in terms of A,

$$B = \frac{k-\kappa}{k+\kappa}A, \qquad C = \frac{2k}{k+\kappa}A. \tag{2.6.4}$$

Now that the speed of the particle is different in the two regions we must use the probability current to calculate the reflection and transmission coefficients. The probability currents for incident, reflected, and transmitted waves are

$$J_{inc} = \frac{\hbar k}{m}|A|^2, \quad J_{ref} = \frac{\hbar k}{m}|B|^2, \qquad J_{tra} = \frac{\hbar \kappa}{m}|C|^2. \tag{2.6.5}$$

The reflection and transmission coefficients are

$$\begin{aligned} R = \frac{J_{ref}}{J_{inc}} &= \left|\frac{B}{A}\right|^2 = \left(\frac{k-\kappa}{k+\kappa}\right)^2 = \left(\frac{\sqrt{E}-\sqrt{E-V_0}}{\sqrt{E}+\sqrt{E-V_0}}\right)^2 \\ T = \frac{J_{tra}}{J_{inc}} &= \frac{\kappa}{k}\left|\frac{C}{A}\right|^2 = \frac{4k\kappa}{(k+\kappa)^2} = \frac{4\sqrt{E}\sqrt{E-V_0}}{(\sqrt{E}+\sqrt{E-V_0})^2} \end{aligned} \tag{2.6.6}$$

Observe that

$$R + T = 1, \tag{2.6.7}$$

consistent with conservation of probability. The coefficients and their sum as a function of energy are shown in the figure below.

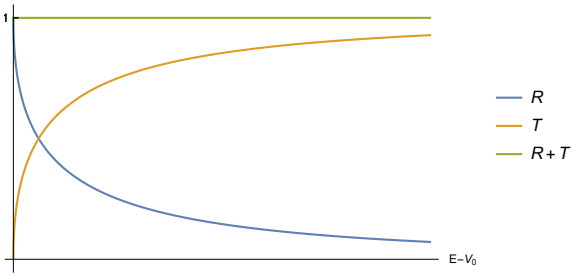


Figure 2.6.2: Reflection and transmission coefficients and their sum for a particle with energy above a step potential.

Question 2.6.1: Compare the reflection or transmission of the quantum particle discussed above with a classical particle.

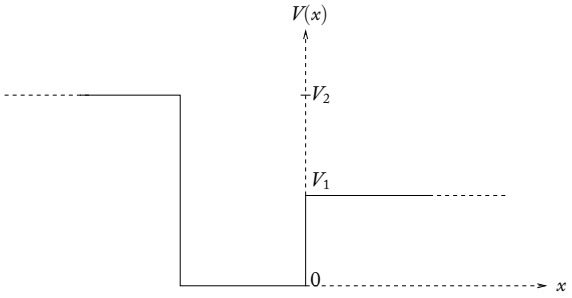
Answer: For a classical particle $R = 1$ and $T = 0$ for $E < V_0$ while $R = 0$ and $T = 1$ for $E > V_0$.

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Question 2.6.2: For $E < V_0$ in the step potential, (A). $B = A$, (B). $B = -A$, (C). $T = 1$, (D). $R = 1$,

Answer: The answer is (D). $R = 1$, since the wavefunction in region II is real, the wave is simply exponentially attenuated and the corresponding transmitted current is zero, and $R = \left|\frac{B}{A}\right|^2 = 1$. To see the math, note that $\psi_I(x) = Ae^{ikx} + Be^{-ikx}$ and $\psi_{II}(x) = Ce^{-\kappa x}$, where $k = \sqrt{\frac{2mE}{\hbar^2}}$ and $\kappa = \sqrt{\frac{2m(V_0-E)}{\hbar^2}}$. Applying the boundary conditions, $A+B = C$ and $ik(A-B) = -\kappa C$, which give $B = -\frac{\kappa+ik}{\kappa-ik}A$ and $C = -\frac{2ik}{\kappa-ik}A$, which give $R = 1$ and $T = 0$.

Question 2.6.3: Which of the following statements are true about the energies of a particle in the potential shown below?



(A). The energies are discrete for $0 < E < V_1$. (B). The energies are discrete for $V_1 < E < V_2$. (C). The energies are discrete for $E > V_2$. (D). The energies are continuous for $0 < E < V_1$. (E). The energies are continuous for $V_1 < E < V_2$. (F). The energies are continuous for $E > V_2$.

Answer: The answers are (A). The energies are discrete for $0 < E < V_1$, (E). The energies are continuous for $V_1 < E < V_2$ and (F). The energies are continuous for $E > V_2$. For $0 < E < V_1$, the wavefunction is constrained by two boundary conditions and the particle is bound in the well and only discrete energies are allowed. There is only one boundary condition on the wavefunction and the energy is continuous for $E > V_1$.

2.7 The simple harmonic oscillator

Now we want to investigate a particle of mass m in the simple harmonic potential

$$V(x) = \frac{1}{2}kx^2 = \frac{1}{2}m\omega^2x^2. \tag{2.7.1}$$

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Because the potential is time-independent, we start with solving the time-independent Schrödinger equation,

$$-\frac{\hbar^2}{2m}\frac{d^2\psi(x)}{dx^2}+\frac{1}{2}m\omega^2x^2\psi(x)=E\psi(x). \tag{2.7.2}$$

Because the potential becomes infinite at $x=\pm\infty$, the particle cannot be found there, since it would need infinite energy. Alternatively, because the wavefunction needs to be normalizable, it has to vanish at $x=\pm\infty$. Therefore the boundary conditions are

$$\psi(-\infty)=0, \qquad \psi(+\infty)=0. \tag{2.7.3}$$

The wavefunction starts with $\psi(-\infty)=0$ at $x=-\infty$, oscillates according to the Schrödinger equation in the interior, and falls back to $\psi(+\infty)=0$ at $x=+\infty$. This equation is not easy to solve immediately. Therefore, we follow a number of steps.

Step 1: Simplify notation with change of variable

The first step is to rewrite the equation in simpler form without too many constants floating around. To do that, we make a change of variable of the coordinate x and define

$$\xi=\sqrt{\frac{m\omega}{\hbar}}x. \tag{2.7.4}$$

We then have $\frac{d}{dx}=\frac{d\xi}{dx}\frac{d}{d\xi}=\sqrt{\frac{m\omega}{\hbar}}\frac{d}{d\xi}$ and $\frac{d^2}{dx^2}=\frac{m\omega}{\hbar}\frac{d^2}{d\xi^2}$. The Schrödinger equation then takes the form

$$\frac{d^2\psi(\xi)}{d\xi^2}=(\xi^2-K)\psi(\xi), \tag{2.7.5}$$

where

$$K\equiv\frac{2E}{\hbar\omega}. \tag{2.7.6}$$

Step 2: Asymptotic behavior

If we have an approximate solution in some region of ξ, then we can write a solution ansatz that is a product of the approximate solution and an unknown function $H(\xi)$ that would allow to solve the general equation exactly. Hopefully, the equation for $H(\xi)$ is easier to solve. In particular, (2.7.5) is difficult because of the ξ^2 term. If we could have approximate solution that solves it at this order, then this term would be eliminated. Because the ξ^2 term dominates over the K term for large ξ, let us work in the large $\xi^2 >> K$ region, where the equation reduces

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to

$$\frac{d^2\psi(\xi)}{d\xi^2} \sim \xi^2\psi(\xi). \tag{2.7.7}$$

Now we can immediately write approximate solutions,

$$\psi(\xi) \sim e^{-\frac{1}{2}\xi^2}, \qquad \psi(\xi) \sim e^{\frac{1}{2}\xi^2}. \tag{2.7.8}$$

Because a solution that goes like $e^{\frac{1}{2}\xi^2}$ at large ξ cannot be normalizable, we discard it and keep only $\psi(\xi) \sim e^{-\frac{1}{2}\xi^2}$. Note that this gives $\psi''(\xi) \sim \xi^2 e^{-\frac{1}{2}\xi^2} - e^{-\frac{1}{2}\xi^2}$ which approximately equals $\xi^2 e^{-\frac{1}{2}\xi^2}$, since it dominates over $-e^{-\frac{1}{2}\xi^2}$ for large ξ, and thus the approximate equation is approximately solved. That is all well, since we are not seeking exact solutions at this point. We then write an ansatz solution,

$$\psi(\xi) = H(\xi)e^{-\frac{1}{2}\xi^2}; \tag{2.7.9}$$

we want to solve for $H(\xi)$ exactly and produce the exact $\psi(\xi)$. Differentiating this,

$$\frac{d^2\psi(\xi)}{d\xi^2} = H''(\xi)e^{-\frac{1}{2}\xi^2} - 2\xi H'(\xi)e^{-\frac{1}{2}\xi^2} + H(\xi)e^{-\frac{1}{2}\xi^2} + \xi^2 H(\xi)e^{-\frac{1}{2}\xi^2}. \tag{2.7.10}$$

Putting (2.7.9) and (2.7.10) in (2.7.5), we have

$$H''(\xi)e^{-\frac{1}{2}\xi^2} - 2\xi H'(\xi)e^{-\frac{1}{2}\xi^2} + H(\xi)e^{-\frac{1}{2}\xi^2} + \cancel{\xi^2 H(\xi)e^{-\frac{1}{2}\xi^2}} = \cancel{\xi^2 H(\xi)e^{-\frac{1}{2}\xi^2}} - KH(\xi)e^{-\frac{1}{2}\xi^2}. \tag{2.7.11}$$

Observe that the ξ^2 terms cancel out, since the approximate solution $e^{-\frac{1}{2}\xi^2}$ solves the Schrödinger equation at leading order in large ξ.

Thus, the Schrödinger equation reduces to an equation for $H(\xi)$.

$$H''(\xi) - 2\xi H'(\xi) + (K-1)H(\xi) = 0. \tag{2.7.12}$$

This equation is called the Hermite equation.

Step 3: Method of power series

It is known in the theory of differential equations that if the equation is written with the coefficient of the highest order derivative set to one as in here, then the equation could be singular only at the singular points of the coefficients of the lower order derivative terms. The coefficients we have in (2.7.12) have no singularity and, therefore, $H(\xi)$ can have no singularities at finite ξ and a power series solution will have an infinite radius of convergence. Thus, we can

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write a power series solution of the form

$$H(\xi) = \sum_{n=0}^{\infty} a_n \xi^n. \tag{2.7.13}$$

The idea is then to rewrite (2.7.12) as $\sum[\ldots]_n \xi^n = 0$, with constant coefficients $[\ldots]_n$, which holds for all ξ only if each constant coefficient vanishes; i.e., $[\ldots]_n = 0$ for each n. Therefore, we seek to write each term in (2.7.12) as $\sum[\ldots]_n \xi^n = 0$ so that we can combine the terms together. First, $H'(\xi) = \sum_{n=0}^{\infty} n a_n \xi^{n-1}$. Because what we have in (2.7.12) is $-2\xi H'(\xi) = -2\sum_{n=0}^{\infty} n a_n \xi^n$, which has the desired form, we keep it as is. Next, taking the second derivative, $H''(\xi) = \sum_{n=0}^{\infty} n(n-1) a_n \xi^{n-2}$. Now this has ξ^{n-2} while we seek ξ^n so that we can combine the coefficients with the other terms. But the first two terms for $n = 0$ and $n = 1$ are zero, and we can start the sum with $n = 2$, $H''(\xi) = \sum_{n=2}^{\infty} n(n-1) a_n \xi^{n-2}$. Next we shift $n \to n+2$ and the sum starts at $n = 0$ as in the others, which we do by rewriting it as $H''(\xi) = \sum_{n=0}^{\infty} (n+2)(n+1) a_{n+2} \xi^n$, and we have expanded it in terms of ξ^n too. Putting these in (2.7.12), we have

$$\sum_{n=0}^{\infty} \left[(n+2)(n+1) a_{n+2} - (2n+1-K) a_n\right] \xi^n = 0. \tag{2.7.14}$$

This holds for all ξ only if each and every coefficient of ξ^n is identically zero, which gives the recursion relation

$$a_{n+2} = \frac{2n+1-K}{(n+2)(n+1)} a_n. \tag{2.7.15}$$

Observe that there are only two independent coefficients, since starting with a_0, we recursively produce a_2, a_4, etc, which produces even solutions. If we start with a_1, we recursively obtain a_3, a_5, etc, which produces odd solutions.

Step 4: Infinite series produces a diverging function

For large n, we have

$$\frac{a_{n+2}}{a_n} \to \frac{2}{n} \tag{2.7.16}$$

But this is similar to the behavior in the series expansion of $\xi^m e^{\xi^2}$ for any m, which goes like e^{ξ^2} for large ξ. In other words, if we keep all infinite series of terms, then we would have $H(\xi) \sim e^{\xi^2}$ and $\psi(\xi) = H(\xi) e^{-\frac{1}{2}\xi^2} \sim e^{\frac{1}{2}\xi^2}$, and the wavefunction would blow up at large ξ and would not be normalizable.

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Step 5: Truncate series

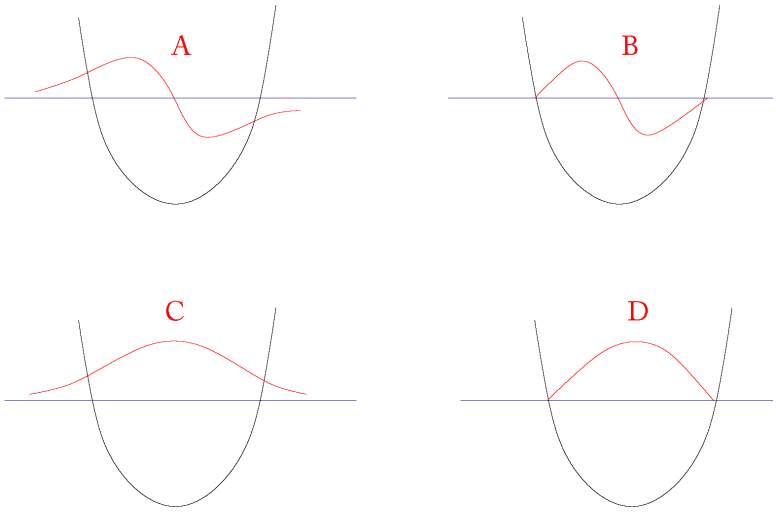
We need to truncate the sum so that $H(\xi)$ would at most be a polynomial of some finite degree n, and then $\psi(\xi) = H(\xi)e^{-\frac{1}{2}\xi^2}$ would be finite, since $e^{-\frac{1}{2}\xi^2}$ decays much stronger than any polynomial of finite degree at large ξ. We truncate the series at n by setting $a_{n+2} = 0$ while $a_n \neq 0$, which is achieved when $2n + 1 - K = 0$, or using the definition of K,

$$E_n = (n + \frac{1}{2})\hbar\omega, \tag{2.7.17}$$

where we could choose to truncate the series at $n = 0$, $n = 1$, $n = 2$, etc, which produce polynomials of the same degree. These energies are the energy levels of the simple harmonic oscillator, and the corresponding polynomials are the Hermite polynomials (up to irrelevant overall constant factors).

2.7.1 Simple harmonic oscillator wavefunctions

Question 2.7.1: Which one of the following best resembles the first excited state of the simple harmonic oscillator?



Answer: The answer is (A), since the first exited state would have to cross the $x-$axis once, in contrast to the ground state (represented by (C)) which does not. Moreover, the wavefunction does not vanish right outside in the classically forbidden region, since the potential is finite at finite x outside the classical turning points until one goes to $\pm\infty$ where the wavefunction vanishes.

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Let us start with truncating the series at $n = 0$ such that $a_0 \neq 0$ while $a_2 = 0$. Because $a_2 = \frac{2n+1-K}{(n+2)(n+1)} a_n|_{n=0} = \frac{1-K}{2} a_0$, we set $1 - K = 0$ which gives

$$E_0 = \frac{1}{2}\hbar\omega. \tag{2.7.18}$$

We also have $H_0(\xi) = a_0$, and the wavefunction is $\psi_0(\xi) = H_0(\xi)e^{-\frac{1}{2}\xi^2} = a_0 e^{-\frac{1}{2}\xi^2} = a_0 e^{-\frac{m\omega}{2\hbar}x^2}$. Normalizing, the ground state wavefunction is

$$\psi_0(x) = \left(\frac{m\omega}{\pi\hbar}\right)^{1/4} e^{-\frac{m\omega}{2\hbar}x^2}. \tag{2.7.19}$$

To construct the first excited state, we truncate the series at $n = 1$ such that $a_3 = 0$ while $a_1 \neq 0$. Therefore, $a_3 = \frac{2n+1-K}{(n+2)(n+1)} a_n|_{n=1} = \frac{3-K}{6} a_1 = 0$, or $3 - K = 0$, which gives

$$E_0 = \frac{3}{2}\hbar\omega. \tag{2.7.20}$$

Furthermore, $H_1(\xi) = a_1\xi$, and the wavefunction is $\psi_1(\xi) = H_1(\xi)e^{-\frac{1}{2}\xi^2} = a_1\xi e^{-\frac{1}{2}\xi^2} = a_1\sqrt{\frac{m\omega}{\hbar}}x e^{-\frac{m\omega}{2\hbar}x^2}$. Normalizing, the first excited state wavefunction is

$$\psi_1(x) = \sqrt{2}\left(\frac{m^3\omega^3}{\pi\hbar^3}\right)^{1/4} x e^{-\frac{m\omega}{2\hbar}x^2}. \tag{2.7.21}$$

The polynomials $H_n(\xi)$ are called the Hermite polynomials (up to an irrelevant overall constant factor).

Now we can continue to construct any of the stationary wavefunctions. Suppose we want to construct $\psi_n(x)$. First, we find K by setting $a_{n+2} = 0$ while $a_n \neq 0$. We then use this K to determine a_2, a_4, ..., a_n in terms of a_0 for even n, or a_3, a_5, ..., a_n in terms of a_1 for odd n. Note that the coefficients are generally different for different states, since the coefficients depend on K (or energy).

The normalized wavefunction $\psi_n(x)$ for the nth state could be conveniently written as

$$\psi_n(\xi) = \left(\frac{m\omega}{\pi\hbar}\right)^{1/4} \frac{1}{\sqrt{2^n n!}} H_n(\xi) e^{-\frac{1}{2}\xi^2}, \tag{2.7.22}$$

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where the Hermite polynomials $H_n(\xi)$ are given by the Rodrigues formula

$$H_n(\xi) = (-1)^n e^{\xi^2} \left(\frac{d}{d\xi}\right)^n e^{-\xi^2} \tag{2.7.23}$$

It can be shown that these wavefunctions form a complete orthonormal set for square-integrable wavefunctions of the harmonic oscillator.

The first few wavefunctions and the corresponding energies are shown in the schematic figure below.

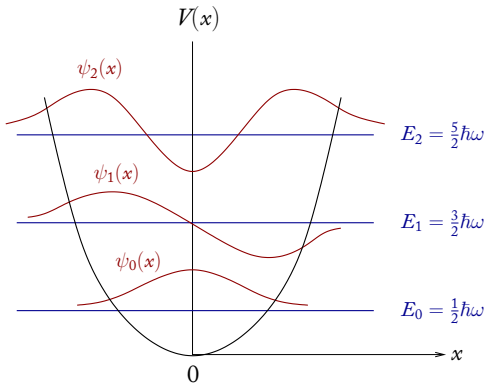


Figure 2.7.1: Wavefunctions and energies for a particle in the simple harmonic oscillator potential.

Question 2.7.2: Discuss why the stationary states for large n are peaked near the boundary of the simple harmonic oscillator as shown in the figure below, in contrast to the low lying states.

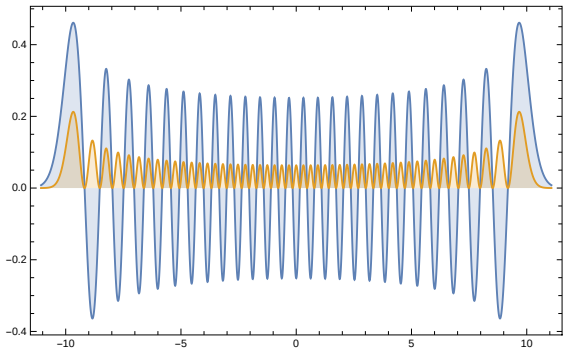


Figure 2.7.2: Wavefunction (blue) and probability density (orange) of the simple harmonic oscillator for large n, $n = 50$ taken here.

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Answer: Larger n corresponds to larger energy. Large energy limit is a classical limit, since a classical oscillator has much larger energy of $\frac{1}{2}m\omega^2A^2 >> \hbar\omega$, where A is the amplitude of oscillation. In this limit the particle behaves more classically. Because a classical oscillator slows down as the particle is near the classical turning points, there is a higher probability for finding it there than in the middle, which is opposite to the low lying states such as the ground state. For large energy E, the energy spacing of $\hbar\omega$ is so small compared to the energy that the energy levels appear continuous as in a classical oscillator. Because $E_n = (n + \frac{1}{2})\hbar\omega$, large n means $n\hbar \approx (n+1)\hbar\omega$. In other words, n being large can be seen as $\hbar$ being small. Thus, we can view the classical limit as the $\hbar \to 0$ limit, which is consistent with our expectation of the energy levels appearing continuous in the classical limit.

Exercise 2.7.1: Show the following Gaussian integrals.

1. $\int_{-\infty}^{\infty} e^{-ax^2}dx = \sqrt{\frac{\pi}{a}}$ and $\int_0^{\infty} e^{-ax^2}dx = \frac{1}{2}\sqrt{\frac{\pi}{a}}$.
2. $\int_{-\infty}^{\infty} x^{2n+1}e^{-ax^2}dx = 0$ for $n = 0, 1, 2, ...$
3. $\int_{-\infty}^{\infty} x^{2n}e^{-ax^2}dx = (-1)^n\frac{d^n}{da^n}\sqrt{\frac{\pi}{a}}$ for $n = 0, 1, 2, ...$
4. $\int_{-\infty}^{\infty} e^{-ax^2+bx}dx = \sqrt{\frac{\pi}{a}}e^{\frac{b^2}{4a}}$.
5. $\int_{-\infty}^{\infty} x^n e^{-ax^2+bx}dx = \frac{d^n}{db^n}\sqrt{\frac{\pi}{a}}e^{\frac{b^2}{4a}}$.

Exercise 2.7.2:

1. Starting with $\psi(\xi) = H(\xi)e^{-\frac{1}{2}\xi^2}$, $H(\xi) = \sum_{n=0}^{\infty} a_n\xi^n$, and the recursion relation $a_{n+2} = \frac{2n+1-K}{(n+2)(n+1)}a_n$ with $\xi \equiv \sqrt{\frac{m\omega}{\hbar}}x$ and $K \equiv \frac{2E}{\hbar\omega}$, determine the energies and the corresponding wavefunctions for the second ($n = 2$) and the third ($n = 3$) excited states of a particle of mass m in a simple harmonic oscillator with potential $V(x) = \frac{1}{2}m\omega^2x^2$.
2. Starting with the Rodrigues formula for $H(\xi)$, $H_n(\xi) = (-1)^n e^{\xi^2}\left(\frac{d}{d\xi}\right)^n e^{-\xi^2}$, determine the wavefunctions for the second ($n = 2$) and the third ($n = 3$) excited states, check that the results are the same as in part (1) (up to irrelevant overall constant factors).
3. Determine the uncertainties for position Δx and momentum Δp for a particle in the ground state of the simple harmonic oscillator, and check whether the uncertainty principle holds.

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4. The particle is in state $\psi(x,0) = \frac{1}{\sqrt{2}}(\psi_0(x) + \psi_1(x))$ of the simple harmonic oscillator at time $t = 0$.

   (a) Write down the wavefunction $\psi(x,t)$ at time t. Set $\hbar = m = \omega = 1$ and use Mathematica to plot $|\psi(x,t)|^2$ at $t = 0$, $t = \frac{\pi}{2}$, $t = \pi$, $t = \frac{3\pi}{2}$, and $t = 2\pi$. (5 points)

   (b) Determine $\langle \hat{x} \rangle$ at time t, and find the amplitude and frequency of oscillation of $\langle \hat{x} \rangle$.

   (c) Determine $\langle \hat{p} \rangle$ at time t.

2.8 Tunneling

Now consider a particle mass m and energy $E < V_0$ approaching from the left a potential barrier of height V_0,

$$V(x) = \begin{cases} V_0, & -a < x < a \\ 0, & |x| > a. \end{cases}$$

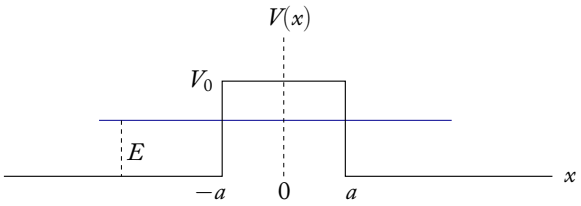


Figure 2.8.1: A particle with energy below the height of a potential barrier.

Denoting the region in $x < -a$ region I, that within $-a \leq x \leq a$ region II, and the region in $x > a$ region III, we can immediately write down the wavefunction in each region,

$$\psi_I(x) = Ae^{ikx} + Be^{-ikx}, \qquad \psi_{II}(x) = Ce^{\kappa x} + De^{-\kappa x}, \qquad \psi_{III}(x) = Fe^{ikx}, \tag{2.8.1}$$

where

$$k = \sqrt{\frac{2mE}{\hbar^2}}, \qquad \kappa = \sqrt{\frac{2m(V_0 - E)}{\hbar^2}} \tag{2.8.2}$$

Applying continuity of $\psi(x)$ and $\psi'(x)$ at $x = \pm a$ produces four equations which can be solved for the four unknowns B, C, D, and F in terms of A. The energy is not an unknown variable

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any continuous incident energy is possible. The solutions are

$$B = \frac{e^{-2ika}\left(e^{4a\kappa}-1\right)\left(k^2+\kappa^2\right)}{e^{4a\kappa}(k+i\kappa)^2-(k-i\kappa)^2}A, \qquad C = -\frac{2k(k-i\kappa)e^{a(\kappa-ik)}}{e^{4a\kappa}(k+i\kappa)^2-(k-i\kappa)^2}A,$$
$$D = \frac{2k(k+i\kappa)e^{3a\kappa-iak}}{e^{4a\kappa}(k+i\kappa)^2-(k-i\kappa)^2}A, \qquad F = \frac{4ik\kappa e^{2a(\kappa-ik)}}{e^{4a\kappa}(k+i\kappa)^2-(k-i\kappa)^2}A. \tag{2.8.3}$$

Our interest here is in the coefficient of transmission T, and the coefficient of reflection would follow from $R = 1 - T$ if needed. Because the speed of the particle in regions I and II is the same, we could simply take the ratio of F and A to find T, and we have

$$T = \left|\frac{F}{A}\right|^2 = \frac{(2k\kappa)^2}{(k^2+\kappa^2)^2\sinh^2(2\kappa a)+(2k\kappa)^2}. \tag{2.8.4}$$

Suppose the barrier's height is much larger than the energy of the particle, $V_0 >> E$. In this case, $\kappa = \sqrt{\frac{2m(V_0-E)}{\hbar^2}}$ and $\sinh^2(2\kappa a) \approx \frac{1}{4}e^{2\sqrt{\frac{2m(V_0-E)}{\hbar^2}}a}$. Because this exponential term dominates over the second term in the denominator in (2.8.4),

$$T \approx \left(\frac{4k\kappa}{k^2+\kappa^2}\right)^2 e^{-2\sqrt{\frac{2m(V_0-E)}{\hbar^2}}2a}. \tag{2.8.5}$$

Taking the natural log of this,

$$\ln T \approx -2\sqrt{\frac{2m(V_0-E)}{\hbar^2}}2a + \ln\left(\frac{4k\kappa}{k^2+\kappa^2}\right)^2 \approx -2\sqrt{\frac{2m(V_0-E)}{\hbar^2}}\,2a, \tag{2.8.6}$$

where we ignored the second term because the logarithm of a number is much smaller than the number. Therefore,

$$T \approx e^{-2\sqrt{\frac{2m(V_0-E)}{\hbar^2}}2a}, \tag{2.8.7}$$

Note that $T \to 0$ as $a \to \infty$,

Next let us consider a particle incident with energy much smaller than the height of a barrier of general shape that is not necessarily a square one as shown in the figure below.

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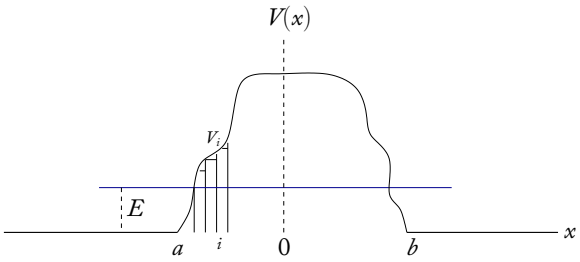


Figure 2.8.2: A particle incident on a potential barrier of general shape.

We can divide the barrier into infinitesimal square barriers each of width Δx, and assuming that the height of each infinitesimal barrier is much bigger than the energy E of the particle, we can start with (2.8.6) for each subdivision, and adding them

$$\ln T = \sum_i \ln T_i \approx -2 \sum_i \sqrt{\frac{2m(V_i - E)}{\hbar^2}}\, \Delta x, \tag{2.8.8}$$

where V_i is the height of the potential at the ith division as shown in the figure above. Taking the limit $\Delta x \to 0$, the sum becomes an integral, and we have

$$\ln T \approx -2 \int_a^b \sqrt{\frac{2m(V(x) - E)}{\hbar^2}}\, dx, \tag{2.8.9}$$

or

$$T \approx e^{-2\int_a^b \sqrt{\frac{2m(V(x)-E)}{\hbar^2}}\, dx} = e^{-2\gamma}, \tag{2.8.10}$$

where

$$\gamma = \int_a^b \sqrt{\frac{2m(V(x) - E)}{\hbar^2}}\, dx \tag{2.8.11}$$

This is the tunneling formula for calculating the coefficient of transmission across a barrier potential.

Question 2.8.1: Why is a particle allowed to be found in classically forbidden region where $E < V$?

Answer: That is because the particle is represented by a wave in quantum mechanics and it is spread out with nonzero probability effectively everywhere (though the probability could be very small). The infinite square well with the wavefunction vanishing outside the well is an ideal mathematical abstraction, and there is no actually infinite potential and the wavefunction would be smoothly going to zero

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at infinity outside the well in reality.

An alternative viewpoint makes use of the uncertainty in position and momentum $\Delta x\Delta p \geq \frac{\hbar}{2}$, which implies $\Delta x\Delta p = \frac{\Delta x}{\Delta t}\Delta p\Delta t = v\Delta p\Delta t = \Delta E\Delta t \geq \frac{\hbar}{2}$, where we have considered a free particle with $\Delta E = \Delta\frac{p^2}{2m} = \frac{p}{m}\Delta p = v\Delta p$. In general, the uncertainty in energy ΔE is related to Δt and

$$\Delta E\Delta t \geq \frac{\hbar}{2}. \tag{2.8.12}$$

Therefore, $\Delta E \geq \frac{\hbar}{2\Delta t}$, and the particle can have as much energy as possible for a short time Δt. However, the standard picture of tunneling is just the nonvanishing wavefunction inside a potential barrier that connects two classically allowed regions.

Exercise 2.8.1: Estimate the lifetime of a particle of mass m and energy E in the potential well shown in the figure below.

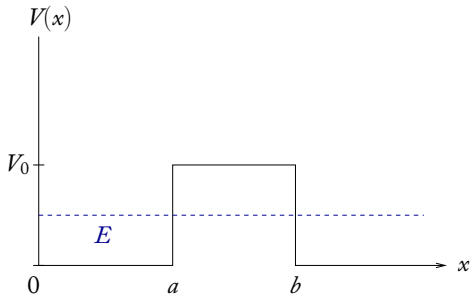


Figure 2.8.3: Particle in a potential well of finite barrier on the right.

Answer: The particle has velocity $v = \sqrt{\frac{2E}{m}}$ and bounces back and forth in the well. Whenever it hits the right wall, it has a probability $T = e^{-2\gamma}$ for tunneling, where $\gamma = \sqrt{\frac{2m(V_0-E)}{\hbar^2}}(b-a)$. It needs a number N of bounces on the right wall such that $NT \sim 1$ for it to tunnel. Therefore, $N \sim \frac{1}{T} = e^{2\gamma}$. On the other hand the time interval between bounces off the right wall is $t = \frac{2a}{v}$. The lifetime is $\tau \sim Nt = \frac{2a}{\sqrt{\frac{2E}{m}}}e^{2\sqrt{\frac{2m(V_0-E)}{\hbar^2}}(b-a)}$.

Question 2.8.2: A scanning tunneling microscope is used to produce an image of the surface of materials at atomic level. A schematic diagram of how the microscope is used is shown in the figure below for a material with lattice spacing of a between the atoms. A sharp tungsten wire tip (of *nano* scale size thickness) is

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brought close to the surface of a conducting material. A piezoelectric scanner body above the tip expands and contracts with applied voltage, and controls the distance from the material. The electrons at the tip of the tungsten wire tunnel across the gap into the material and produce a current depending on the position of the tip relative to the atoms in the material.

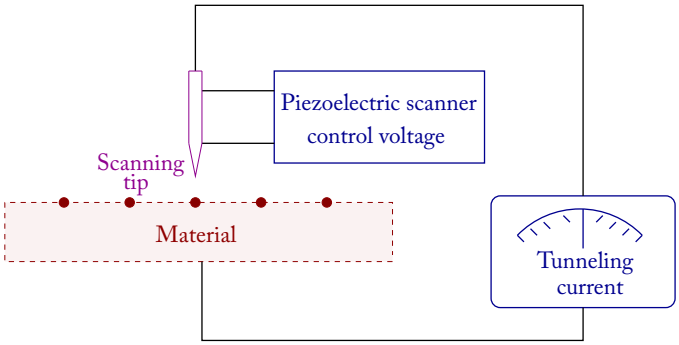


Figure 2.8.4: A schematic description of the scanning tunneling microscope.

This current is then amplified and displayed. Using the magnitude of the current, one can then construct an image of the shape of the local density of states (corresponding to the density of electrons) on the surface of the material. An example of such an image is shown below (taken from the International Business Machines (IBM) Corporation).

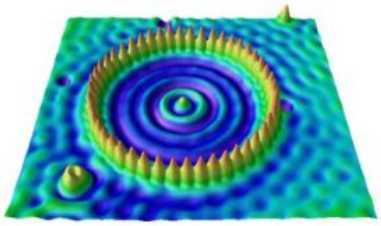
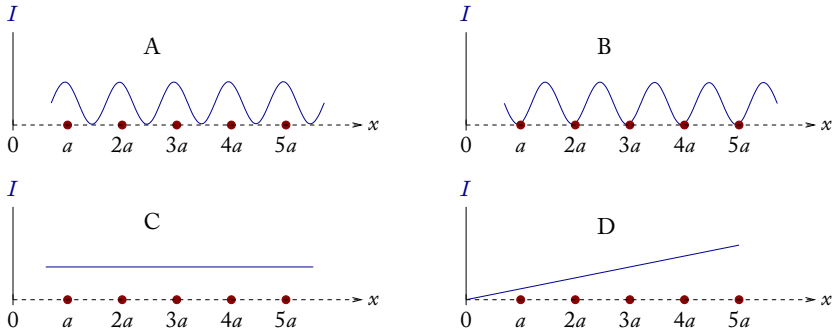


Figure 2.8.5: A scanning tunneling microscope image from IBM.

Which of the following figures best represents the current measured as a function of position across a conducting surface, at a fixed tip distance from the surface?

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Answer: The answer is A. An electron in the sharp tip of the microscope's can be viewed as an electron in a potential well. The atomic protons in the nucleus of an atom in the material provide another deeper potential well for electrons. The air gap between the tip and the surface provides a potential barrier. When the tip is closer and just above an atom, the potential barrier becomes smaller (the distance between the tip and the surface is smaller too, and the surface is like valleys and hills at atomic scale) and electrons in the tip tunnel through, producing a tunneling current, as schematically shown in the figure below. In particular, the

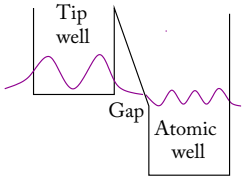


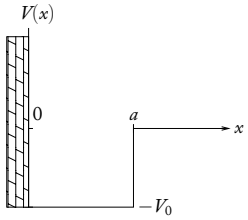
Figure 2.8.6: Schematic wells and barriers in the electron tunneling microscope scheme.

The current is then amplified. Amplification is necessary since the typical current measured is in the order of *nanoamperes*. The magnitude of measured current provides a measure of the density of states in the material or the structure of the material. In other words, the current oscillation mimics the structure (or the atomic distribution at the surface) of the material.

Exercise 2.8.2:

1. Consider a particle of mass m in one dimension in the potential shown in the figure below with $V(x) = \infty$ for $x < 0$, $V(x) = -V_0$ for $0 \leq x \leq a$ and $V(x) = 0$ for $x > a$.

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(a) Determine the transcendental equation for the energy of bound states in the well.

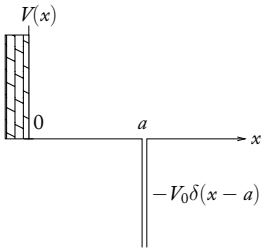
(b) Determine the minimum value of V_0 which supports at least one bound state.

(c) Make a schematic drawing of the ground bound state wavefunction over the given potential.

(d) Determine the wavefunction for a particle with energy $E > 0$ in each region $0 \leq x \leq a$ and $x > a$.

(e) Determine the phase difference between the wavefunction in the region $x > a$ and the wavefunction in $0 \leq x \leq a$ for the particle in part (1d). What is the source for the phase difference?

2. Next, the potential is changed to the one shown in the figure below where $V(x) = -V_0\delta(x-a)$ for $x > 0$ and a wall with $V(x) = \infty$ for $x < 0$. This potential could be used as an approximation for an atom near a wall.

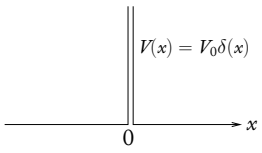


(a) Find a transcendental equation for the energy of a bound state. Determine the condition for which the particle is very far from the wall at $x = 0$ that the wall could be ignored.

(b) Determine the minimum value of V_0 that supports a bound state, in terms of other given parameters.

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3. Next, the potential is changed to the one shown in the figure below with $V(x) = V_0\delta(x)$. A particle of energy $E > 0$ is incident from the left toward the origin.



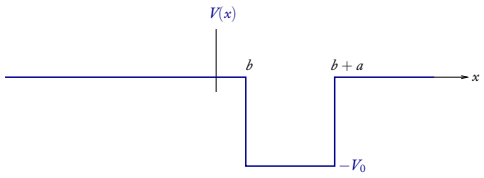
   (a) Determine the ratio of the amplitudes of the wave transmitted to the right to that of the incident and the ratio of the amplitudes of the wave reflected back to the left to that of the incident.

   (b) Make schematic plots of the transmission and the reflection coefficients and their sum as a function of energy E. Do the plots make sense? (5 points)

Exercise 2.8.3:

1. Single and double potential wells. Answer these questions qualitatively without doing calculations.

   (a) Make a schematic plot of the ground state wavefunction for a bound particle in the single potential well of width a and depth V_0 below. Denote the wavefunction by $\psi(x)$ and the energy by E.



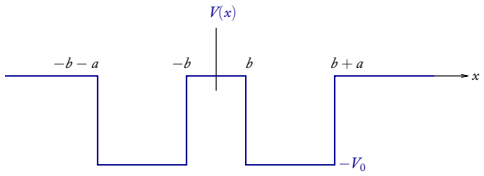
   (b) Make a schematic plot of $E + V_0$ as a function of V_0, from $V_0 = 0$ to $V_0 = \infty$, with a fixed. Mark the values at $V_0 = 0$ to $V_0 = \infty$.

   (c) Make a schematic plot of $E + V_0$ as a function of a, from $a \approx 0$ to $a = \infty$, with V_0 fixed.

   (d) Make a schematic plot of the ground state and the first exited state wavefunctions, denote them by $\psi_s(x)$ and $\psi_a(x)$, for a bound particle in the double potential well below. Assume that the wells are deep enough

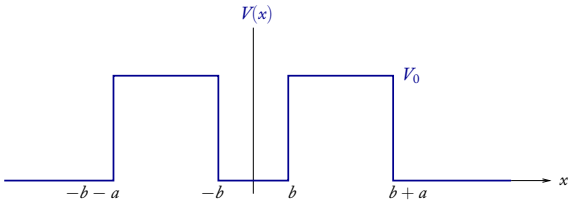
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that each well can be separately viewed as close to (but not exactly) an infinite square well for the lower lying states. Point out how $\psi_s(x)$ and $\psi_a(x)$ are qualitatively related to the $\psi(x)$ of a similarly deep well in part (1a). Would the corresponding energies E_s and E_a be equal, greater, or smaller than E of the single well. (5 points) [Double well potentials are used for various physical systems such as H_2 molecule.]



(e) Make a schematic plot of both $E_s + V_0$ and $E_a + V_0$ as a function of b, from $b = 0$ to $b = \infty$, and point out how these are related to E.

2. Now the double potential well is inverted upside down and converted to the double potential barrier shown in the figure below.



(a) Estimate the lifetime τ of a bound particle of mass m and energy E between the barriers in terms of given parameters. [Assume that E and the depth and the width of the barriers are such that the tunneling formula applies.]

(b) Make schematic plots of τ as a function of V_0, from $V_0 = 0$ to $V_0 = \infty$, with a fixed and $E \approx 0$, and also as a function of a, from $a = 0$ to $a = \infty$, with V_0 fixed.

2.9 Optional section on application of tunneling: Gamow's theory of $\alpha-$decay

We want to model radioactive $\alpha-$decay as a quantum mechanical tunnelling process and apply the tunneling formula we have obtained to estimate the lifetime of heavy nuclei. First let us

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recall features in the binding energy per nucleon of nuclei that are relevant to our discussion, which we have schematically plotted below.

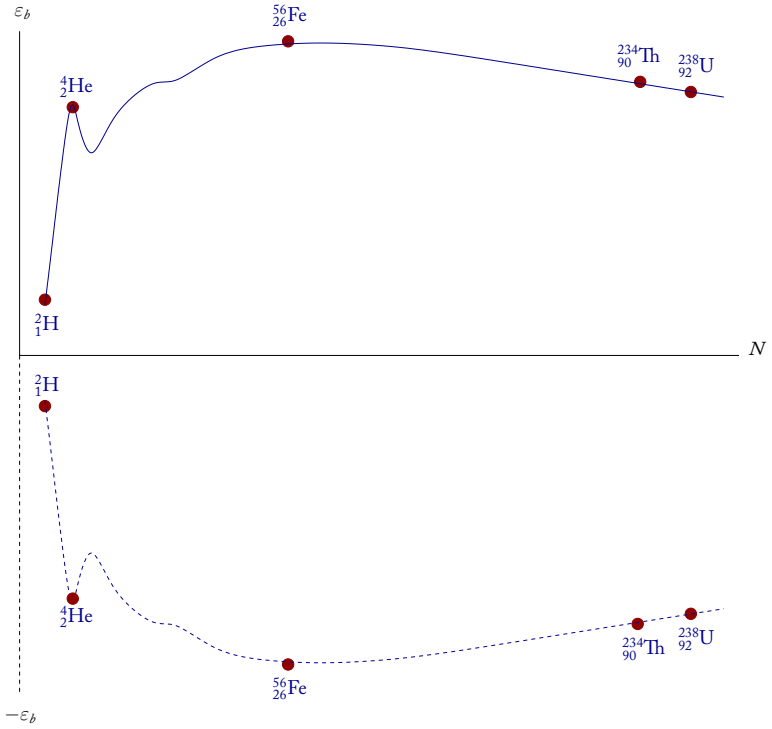


Figure 2.9.1: Schematic plot of binding energy per nucleon (ε_b) versus nucleon number (N). The solid line represents the magnitude and the dashed line is for the actual negative energy.

The main feature for us here from the above plot is that the binding energy per nuclei decreases with increasing nucleon number in the region of heavy radioactive nuclei such as $^{238}_{92}$U. Note that the actual energy per nucleon for bound nucleons is negative as represented by the dashed line, and a heavy parent nucleus decays to a lighter daughter nucleus by releasing an alpha particle of positive kinetic energy. As an example, a nucleon in $^{238}_{92}U$ is less tightly bound than a nucleon in $^{234}_{90}Th$. An $\alpha-$particle is the nucleus of $^{4}_{2}$He atom, and its binding energy per nucleon is relatively larger in comparison to other nuclei of comparable nucleon number. Inside heavy nuclei such as $^{238}_{92}$U, we can have a bound $\alpha-$particle of positive energy which behaves very much like a free particle bouncing inside the heavy nucleus. The strong nuclear interaction has finite range and operates within nuclear size of order $10^{-15}m$. We can imagine the interaction of the $\alpha-$particle with the rest of a heavy nucleus as mediated by a string with the $\alpha-$particle and the rest of the heavy nucleus at the ends of the string. When the two are very close within the nucleus, the string is slack and the $\alpha-$particle is very much a free particle. When the $\alpha-$particle and the rest of the nucleus are separated by distance of $10^{-15}m$ the string becomes very tight and exerts a

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large force on it. Just outside this range of distance, the $\alpha-$particle interacts with the rest of the nucleus through repulsive Coulomb force, since the particle is outside operating range of the nuclear force. The $\alpha-$decay corresponds to a tunneling of the $\alpha-$particle through this potential. Considering a heavy nucleus of atomic number $Z+2$ and nucleon number $N+4$, the decay process is

$$^{N+4}_{Z+2}X \rightarrow ^{N}_{Z}Y + \alpha, \tag{2.9.1}$$

where X is the parent nucleus and Y is the daughter. The figure below shows a schematic plot of the potential energy of the $\alpha-$particle as a function of the distance from the center of a heavy nucleus.

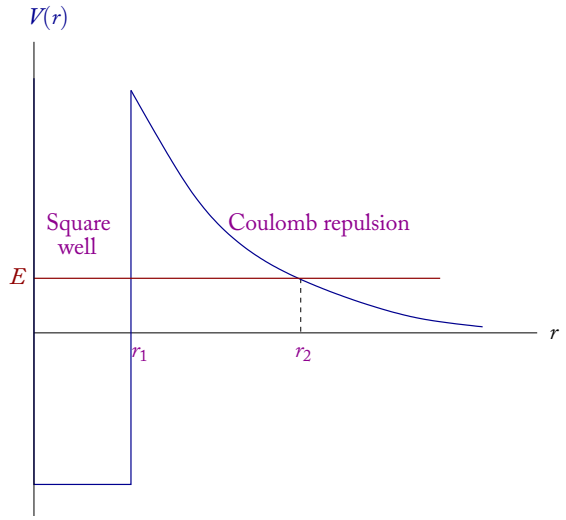


Figure 2.9.2: Schematic plot of the potential energy of $\alpha-$particle in $\alpha-$decay.

Exercise 2.9.1: Calculate the energy of the alpha particle that results from the decay $^{238}_{92}U \rightarrow ^{234}_{90}Th + \alpha$ and point out the binding energy per nucleon in the nuclei involved.

Solution: The masses are $M_U = 238.0508amu$ for $^{238}_{92}U$, $M_{Th} = 234.0436amu$ for $^{230}_{90}Th$, $M_\alpha = 4.00260amu$ for $^{4}_{2}He$, $m_n = 1.00786amu$ for one nucleon, where $1amu = 1.66054 \times 10^{-27}kg$. For a nuclues with N nucleons, the total energy is $M_Nc^2 - N\epsilon_b$, where the first term is for the total mass of the nucleus and the second term is negative and written the negative of a positive binding energy with ϵ_b being the binding energy per nucleon (which is taken as positive). Given the total mass of $N(1amu)$ of the N independent nucleons before they are put together in the nucleus, the binding energy is obtained from $N(1amu) = M_Nc^2 - N\epsilon_b$ or $\epsilon_b = \frac{(M_Nc^2 - N(1amu))}{N}$. The kinetic energy released in the alpha particle is simply the difference in the

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masses and $E_\alpha = (M_U - M_{Th} - M_\alpha)c^2 = 6.87 \times 10^{-13}J = 4.28MeV$. The binding energy per nucleon is $\epsilon_b^U = (\frac{238.0508-238}{238})1.00786amu = 0.000215123amu = 3.21054 \times 10^{-14}J = 0.200386 \times 10^6 eV$ for ${}^{238}_{92}U$, $\epsilon_b^{Th} = (\frac{234.0436-234}{234})1.00786amu = 0.000187789amu = 2.8026 \times 10^{-14}J = 0.174925 \times 10^6 eV$ for ${}^{234}_{90}Th$, and $\epsilon_b^\alpha = (\frac{4.00260-4}{4})1.00786amu = 0.000655109amu = 9.77697 \times 10^{-14}J = 0.610231 \times 10^6 eV$ for ${}^4_2He$. For contrast, $\epsilon_b^{Fe} = (\frac{56.4491-56}{56})1.00786amu = -0.0080829amu = 1.20627\times10^{-12}J = 7.52897\times10^6 eV$ for ${}^{56}_{26}Fe$, and the actual energy per nucleon $-\epsilon_b^{Fe}$ is more negative than for other nuclei and the nucleons in iron are most bound.

The potential is

$$V(r) = \begin{cases} -V_0, & 0 < r < r_1 \\ \frac{2Ze^2}{r}, & r > r_1. \end{cases}$$

The negative constant potential in the region $r < r_1$ is because the nuclear force inside is attractive and binding, with the attractive force becoming very strong at the boundary of the nucleus. The nuclear force operates only within the nucleus and the $\alpha-$particle interacts with the rest of the nucleus through electric interaction outside the nucleus. An $\alpha-$particle with positive energy $E > 0$ can tunnel through the potential barrier as shown in the figure above. Because the strong nuclear force at $r = r_1$ (and the Coulomb force at the same distance) is large, the width of the barrier is such that $r_2 >> r_1$. For a given energy E, the alpha particle moves with speed $v = \sqrt{\frac{2E}{m}}$, where m is its mass, and oscillates within the well with period

$$t = \frac{2r_1}{v}. \tag{2.9.2}$$

The lifetime of the heavy nucleus (the time it takes to decay) can be estimates as follows. For each bounce on the wall on the right at $r = r_1$, the alpha particle has a probability of T for tunneling. It needs N number of bounces such that $NT = O(1)$, which we simply set as 1, $NT = 1$. Therefore, the lifetime is $\tau = Nt = \frac{t}{T}$, or

$$\tau = te^{2\gamma} = \frac{2r_1}{v}e^{2\gamma}. \tag{2.9.3}$$

But

$$\gamma = \frac{1}{\hbar}\int_{r_1}^{r_2} \sqrt{2m(V(r) - E)}\, dr = \frac{1}{\hbar}\int_{r_1}^{r_2} \sqrt{2m(\frac{2Ze^2}{r} - E)}\, dr$$

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$$= \frac{\sqrt{2mE}}{\hbar}\left(r_2 \cos^{-1}(\sqrt{\frac{r_1}{r_2}}) - \sqrt{(r_2 - r_1)r_1}\right) \tag{2.9.4}$$

Because, $r_2 >> r_1$, $\cos^{-1}(\sqrt{\frac{r_1}{r_2}}) \approx \frac{\pi}{2} - \frac{r_1}{r_2}$ and $(r_2 - r_1)r_1 \approx r_1 r_2$, which with $V(r_2) = E$ or $r_2 = \frac{2Ze^2}{E}$ gives

$$\begin{aligned}\gamma &\approx \frac{\sqrt{2mE}}{\hbar}\left(\frac{\pi}{2}r_2 - 2\sqrt{r_1 r_2}\right) = \frac{\sqrt{2mE}}{\hbar}\left(\frac{\pi}{2}\frac{2Ze^2}{E} - 2\sqrt{r_1\frac{2Ze^2}{E}}\right) \\ &= \frac{\sqrt{2m}\pi e^2}{\hbar}\frac{Z}{\sqrt{E}} - 4\sqrt{me^2}\sqrt{r_1 Z}. \end{aligned} \tag{2.9.5}$$

Thus,

$$\ln\tau = 2\gamma + \ln(\frac{2r_1}{v}) \approx \frac{\sqrt{2m}\pi e^2}{\hbar}\frac{Z}{\sqrt{E}} - 4\sqrt{me^2}\sqrt{r_1 Z} + \ln(\frac{2r_1}{v}), \tag{2.9.6}$$

and this result reproduces experimental data reasonably well. In particular, data shows similar linear dependence of $\ln\tau$ on $\frac{Z}{\sqrt{E}}$.

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